Programming with Class IV – VI

Coding Questions (LAQ)

1. ***Employee Management System:*** *Suppose a company has different types of employees:*
   - *Lecturer*
   - *Administrator*

   Each employee's salary is calculated differently. Given below is a sample solution where all codes are in one file. Reorganize the codes into separate files. Modify the codes to incorporate Modern C++.

```cpp
#include <iostream>

class Employee {
public:
    virtual void calculateSalary(double basicSalary) {
        std::cout << "Employee Salary: RM"
                  << basicSalary << std::endl;
    }
};

class Lecturer : public Employee {
public:
    void calculateSalary(double basicSalary) override {
        double allowance = 1000;
        std::cout << "Lecturer Salary: RM"
                  << basicSalary + allowance
                  << std::endl;
    }
};

class Administrator : public Employee {
public:
    void calculateSalary(double basicSalary) override {
        double allowance = 500;
        std::cout << "Administrator Salary: RM"
                  << basicSalary + allowance
                  << std::endl;
    }
};

int main() {
    Employee* emp1 = new Lecturer();
    Employee* emp2 = new Administrator();

    emp1->calculateSalary(4000);
    emp2->calculateSalary(4000);

    delete emp1;
    delete emp2;

    return 0;
}
```

2. ***Library Management System:*** *A library would like to manage information about different types of books. All books share common information such as Book ID and Title, but E-Books have additional information related to file size and download format. Using the class diagram provided below, develop a C++ program that demonstrates the concept of inheritance. Write a C++ program that performs the following tasks:*
    a. ***Part A: Base Class:*** *Create a class named* ***Book*** *that contains:*
        i. ***Data Members***
            - *bookId (int)*
            - *title (string)*
        ii. ***Member Functions***
            - *One default constructor*
            - *One overloaded constructor*
            - *Two write accessor (mutator) functions:*
            - *Two read accessor (accessor) functions:*
            - *A function named displayBook() to display the book details*

    Label all parts of the class using appropriate comments.

    b. ***Part B: Derived Class :*** *Create a class named* ***EBook*** *that inherits publicly from the Book class.*
        i. ***Additional Data Members***
            - *fileSize (double)*
            - *format (string)*
        ii. ***Member Functions***
            - *One default constructor*
            - *One overloaded constructor*
            - *Two write accessor functions:*
            - *Two read accessor functions:*
            - *A function named displayEBook() to display all inherited and derived class information*
    c. ***Part C: Main Program:*** *In the main() function:*
        i. *Create* ***one Book object*** *using the default constructor.*
        ii. *Use the write accessor functions to assign values to the object.*
        iii. *Use the read accessor functions to retrieve and display the values.*
        iv. *Create* ***two EBook objects*** *using overloaded constructors.*
        v. *Use the displayEBook() function to display the details of both EBook objects.*